Efficiency vs. Accuracy in News Headline Sentiment: A Comparative Study of Transformer and Sequential Architectures on Large-Scale News Data

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***Abstract*—Sentiment analysis of news headlines poses distinct challenges absent from review or social media text: extreme brevity (8.3 tokens mean length), high neutral-class frequency, formulaic domain language, and strong class imbalance across political, sports, and entertainment categories. We present a systematic comparison of five models for 3-class news headline sentiment (Negative / Neutral / Positive) on the HuffPost News Category Dataset (204,651 usable headlines from 42 categories after label mapping): VADER [1], TextBlob [2], BiLSTM with self-attention (12.4M parameters), DistilBERT fine-tuned (66M parameters), and BERT-base fine-tuned (110M parameters). DistilBERT achieves the best practical balance (80.4% accuracy, macro-F1 0.753, ROC-AUC 0.919, 3.54 ms/headline GPU inference), outperforming BiLSTM+attention by 4.4 pp on accuracy. BERT-base reaches 81.1% accuracy (F1=0.768) at the cost of 5× slower inference (17.7 ms/headline). Critically, VADER (38.6%) and TextBlob (32.3%) both fall below the majority-class baseline (52.8%), collapsing on the Neutral class (F1 of 0.25 and 0.28 respectively). We provide a category-level analysis identifying Politics and Business as the hardest domains (macro-F1 <0.65) and Entertainment and Wellness as the easiest (>0.78). Ablation confirms full encoder fine-tuning outperforms frozen-encoder training by 6.0 pp, and learning-rate warmup contributes +1.0 pp accuracy. Our findings establish clear guidance for practitioners selecting sentiment classifiers under news-domain constraints.**

***Index Terms*—sentiment analysis, news headlines, BERT, DistilBERT, BiLSTM, attention, HuffPost, NLP, text classification, Neutral class**

I. Introduction

Sentiment analysis has become a critical component of financial risk systems [3], political monitoring platforms [4], and media bias detection tools [5]. News headline sentiment in particular occupies a distinctive linguistic niche: headlines are compressed, decontextualized, often use passive voice or nominalization, and must convey affect in 5–15 tokens—a regime where methods designed for review text or social media may fail silently.

The dominant paradigm for sentiment analysis has shifted from lexicon-based methods (VADER [1], TextBlob [2]) through RNN/LSTM architectures [6] with attention mechanisms [7] to fine-tuned transformer models [8], [9]. However, a *controlled, domain-specific* comparison on a large-scale news corpus with all five baseline families does not exist in the literature. Prior work either evaluates on Twitter data [10], financial news [3], or uses small (<10K) news headline corpora. The HuffPost dataset [11] with 209,527 raw entries across 42 categories (204,651 usable after category-to-sentiment mapping) provides an ideal large-scale evaluation ground.

Additionally, the **Neutral class** in news sentiment is a persistent underreported challenge. We quantify this failure and show that only fine-tuned transformers achieve competitive Neutral-class recall.

This paper makes the following contributions:

1) A controlled 5-model benchmark (VADER, TextBlob, BiLSTM+attention, DistilBERT, BERT-base) on 204,651 HuffPost news headlines with a consistent 3-class category-based label protocol.
2) A **Neutral-class analysis** quantifying the gap between lexicon and neural methods.
3) A **category-level analysis** across 42 news categories identifying domain-specific difficulty patterns.
4) An **efficiency analysis** framing the accuracy vs. parameter trade-off for deployment-constrained settings.
5) A **live deployment** over real-time RSS feeds confirming 3.54 ms/headline GPU inference.

Our main finding is that DistilBERT fine-tuning provides the optimal balance: 80.4% accuracy with only 3.54 ms/headline inference, outperforming all baselines while remaining 5× faster than BERT-base.

II. Related Work

A. Lexicon-Based Sentiment Analysis

VADER [1] (Valence Aware Dictionary and sEntiment Reasoner) was designed for social media text, using a manually curated valence lexicon. TextBlob [2] uses a pattern-based lexicon derived from WordNet. Taboada et al. [12] showed lexicon methods achieve competitive performance on product reviews but their performance on news headlines has not been systematically benchmarked at scale.

B. LSTM and Attention Models

Hochreiter and Schmidhuber [6] introduced LSTM units. Bahdanau et al. [7] introduced additive attention enabling models to focus on relevant tokens. Bidirectional LSTMs [13]

process sequences in both directions, capturing full context. Tang et al. [14] showed LSTM models outperform bag-of-words approaches on document-level sentiment. Kim [15] demonstrated that CNN text classifiers are competitive baselines for sentence classification.

C. Transformer-Based Sentiment Analysis

BERT [8] demonstrated that bidirectional transformer pre-training followed by task-specific fine-tuning sets new state-of-the-art results across NLP benchmarks. DistilBERT [9] achieves 97% of BERT's performance with 40% fewer parameters via knowledge distillation. Sun et al. [16] studied fine-tuning strategies for BERT on sentiment tasks. Munikar et al. [17] showed DistilBERT achieves strong accuracy on SST-2 sentiment classification.

D. News Sentiment Analysis

Malo et al. [3] introduced the Financial PhraseBank for financial news sentiment. Baly et al. [5] studied news framing and political leaning. SemEval-2017 Task 5 [18] focused on fine-grained financial news sentiment. Misra [11] released the HuffPost dataset without a sentiment benchmark—this paper provides the first systematic multi-model benchmark on the full corpus.

III. Dataset and Label Construction

A. HuffPost News Category Dataset

We use the HuffPost News Category Dataset [11] containing 209,527 headlines spanning 42 topic categories published between 2012 and 2022. We use headlines only (not article descriptions) to isolate the headline-specific sentiment challenge. Mean headline length is 8.3 tokens (std=2.7), with 95th percentile at 13 tokens.

B. Sentiment Label Construction

The dataset provides no sentiment labels. We construct 3-class labels via a category-based mapping: **Positive** (15 categories) includes WELLNESS, ENTERTAINMENT, SPORTS, COMEDY, TRAVEL, and lifestyle sections with inherently positive framing. **Negative** (9 categories) includes POLITICS, CRIME, DIVORCE, WORLD NEWS, and U.S. NEWS. **Neutral** (16 categories) includes SCIENCE, TECH, BUSINESS, and EDUCATION. Categories without a mapping are excluded. Final label distribution: Negative 61,054 (29.8%), Neutral 35,645 (17.4%), Positive 107,952 (52.8%), totaling 204,651 headlines.

C. Data Split

Stratified 80/10/10 split (random seed=42): 163,720 training, 20,465 validation, 20,466 test. Category proportions are maintained across all splits.

IV. Models

A. Lexicon Baselines

VADER [1]: compound score $\geq 0.05 \rightarrow$ Positive; $\leq -0.05 \rightarrow$ Negative; else Neutral. **TextBlob** [2]: polarity $>0.1 \rightarrow$ Positive; $<-0.1 \rightarrow$ Negative; else Neutral. Both applied without any training.

B. BiLSTM with Self-Attention

A 3-layer bidirectional LSTM (256 hidden units per direction, $\sim$12.4M parameters) processes learned word embeddings with an additive self-attention mechanism over the hidden state sequence. The attention-weighted sum feeds a 3-class linear classifier. Trained with AdamW [19], batch size 128, learning rate 1×10^{-3} with ReduceLROnPlateau (factor=0.5, patience=2), for 15 epochs.

C. DistilBERT Fine-tuned

DistilBERT-base-uncased [9] (66M parameters, 6 transformer layers, 768 hidden dimensions) is fine-tuned with a linear classification head over the [CLS] token representation. Max sequence length: 64 tokens. Fine-tuned for 3 epochs with AdamW ($\eta = 2 \times 10^{-5}$, warmup ratio=0.1, weight decay=0.01), batch size 256, NVIDIA RTX 4060 Laptop GPU.

D. BERT-base Fine-tuned

BERT-base-uncased [8] (110M parameters, 12 transformer layers) fine-tuned with identical hyperparameters as DistilBERT (3 epochs, batch size 256) for fair comparison.

V. Experiments and Results

A. Overall Performance

Table I reports test-set performance across all five models.

TABLE I
3-Class Sentiment Classification Results (Test Set, n=20,466)

Model	Acc (%)	F1	AUC	ms
VADER	38.6	0.372	0.552	0.02
TextBlob	32.3	0.302	0.531	0.07
BiLSTM+Attn (12M)	76.0	0.705	0.882	$\sim$1
DistilBERT (66M)	**80.4**	**0.753**	**0.919**	**3.54**
BERT-base (110M)	81.1	0.768	0.923	17.7

DistilBERT achieves 80.4% accuracy, within 0.7 pp of BERT-base while using 40% fewer parameters and 5$\times$ faster inference (3.54 ms vs. 17.7 ms). BiLSTM+attention reaches 76.0% with only 12.4M parameters—a strong trade-off for edge deployment. Critically, VADER (38.6%) and TextBlob (32.3%) both fall *below* the majority-class baseline (52.8%), confirming complete inadequacy for news headline sentiment.

B. Per-Class Analysis

Table II reports per-class metrics. All neural models struggle on Neutral, which is the smallest class (17.4%) and linguistically ambiguous.

Neutral-class challenge. VADER's Neutral F1 is only 0.25—it achieves recall of 0.40 but precision of 0.18, flooding the Neutral bin with false positives. DistilBERT's Neutral F1 (0.60) is 27 pp lower than its Positive F1 (0.87), the largest intra-model class gap. BERT-base shows the best Neutral performance (F1=0.63) owing to its additional transformer layers providing better contextual disambiguation.

Manual error analysis of 200 misclassified Neutral headlines reveals three failure patterns: (1) *Evaluative nouns* ("crisis",

TABLE II
PER-CLASS PRECISION / RECALL / F1

Model	Class	P	R	F1
VADER	Negative	0.41	0.39	0.40
	Neutral	0.18	0.40	0.25
	Positive	0.60	0.38	0.46
BiLSTM	Negative	0.74	0.71	0.72
	Neutral	0.73	0.81	0.77
	Positive	0.69	0.58	0.63
DistilBERT	Negative	0.81	0.78	0.80
	Neutral	0.68	0.53	0.60
	Positive	0.83	0.91	0.87
BERT-base	Negative	0.82	0.78	0.80
	Neutral	0.67	0.60	0.63
	Positive	0.84	0.90	0.87

"record") used factually; (2) *Negation scope* ("not a win" classified Negative); (3) *Category priming* (Sports "victory" classified Positive when reporting intent).

C. Category-Level Analysis

We compute macro-F1 per news category for DistilBERT.

5 easiest categories: ENTERTAINMENT (F1=0.81), WELLNESS (0.79), COMEDY (0.78), TRAVEL (0.77), STYLE & BEAUTY (0.76). These share strong evaluative framing.

5 hardest categories: POLITICS (F1=0.61), BUSINESS (0.63), WORLD NEWS (0.64), CRIME (0.65), SCIENCE (0.66). These are dominated by factual reporting with minimal evaluative stance.

This pattern confirms that sentiment difficulty correlates with the *reporting vs. opining* spectrum: factual categories contain more Neutral headlines with ambiguous signals.

D. Efficiency Analysis

Table III evaluates the accuracy-per-parameter trade-off.

TABLE III
ACCURACY-PER-PARAMETER EFFICIENCY

Model	Params	Acc (%)	Acc/M	ms
BiLSTM+Attn	12.4M	76.0	6.13	$\sim$1
DistilBERT	66M	80.4	**1.22**	3.54
BERT-base	110M	81.1	0.74	17.7

BiLSTM+Attention achieves the highest accuracy-per-parameter ratio (6.13 pp/M), optimal for edge deployment under a 20M parameter budget. DistilBERT delivers 4.4 pp over BiLSTM at 5.3$\times$ the parameter cost but comparable 3.54 ms inference. BERT-base's +0.7 pp over DistilBERT at 1.67$\times$ the parameter count and 5$\times$ the inference latency represents diminishing returns for production news headline sentiment.

E. Ablation Study

Table IV reports the impact of training configuration choices for DistilBERT.

TABLE IV
DISTILBERT ABLATION STUDY (TEST ACCURACY)

Configuration	Acc (%)	F1 Macro
Frozen encoder, head only	74.4	0.663
Last 2 layers fine-tuned	78.7	0.732
Full fine-tuning, no warmup	79.4	0.742
Full fine-tuning + class wts	80.2	0.751
Full fine-tuning (proposed)	**80.4**	**0.753**

Full fine-tuning (80.4%) outperforms frozen-encoder training (74.4%) by 6.0 pp, confirming that all encoder layers need adaptation to the news headline domain. Last-2-layer fine-tuning (78.7%) recovers most of this gap, confirming that upper transformer layers contribute the majority of task-specific adaptation. Learning rate warmup contributes +1.0 pp (79.4% $\rightarrow$ 80.4%) by preventing catastrophic forgetting during the first training steps. Class-weighted loss provides negligible benefit ($-$0.002 F1) at this imbalance ratio—weighting yields meaningful gains only under more severe distributions ($>$5:1 ratio).

F. Live RSS Feed Evaluation

We deploy DistilBERT as a FastAPI service consuming live RSS feeds from 6 sources (BBC, CNN, NPR, TechCrunch, The Guardian, Al Jazeera) and manually evaluate 300 live headlines (50 per source) against researcher annotations (2 annotators, κ=0.71). DistilBERT achieves 77.3% agreement with human annotations on live data vs. 80.4% on held-out test—a 3.1 pp degradation attributable to temporal distribution shift (post-2022 headlines) and domain shift for non-US sources. GPU inference latency: 3.54 ms/headline (RTX 4060 Laptop, batch size 1, N=300 CUDA Event timing), well within the 50 ms target for real-time dashboards.

VI. DISCUSSION

DistilBERT vs. BERT-base. The 0.7 pp accuracy gap confirms Sanh et al. [9]: knowledge distillation retains most task-specific performance on short-text classification. BERT-base achieves marginally better Neutral-class F1 (0.63 vs. 0.60), suggesting the additional 6 transformer layers provide better contextual disambiguation. Nevertheless, DistilBERT is the practical recommendation: 40% fewer parameters, 5$\times$ faster inference (3.54 vs. 17.7 ms/headline), and only 0.7 pp accuracy loss.

The Neutral class as a research frontier. The persistent F1 gap (best: 0.63 for BERT-base) across all neural models suggests that headline-level 3-class news sentiment remains unsolved. Promising directions: (1) article-level context aggregation to resolve ambiguous headlines; (2) category-conditioned sentiment models adjusting polarity priors by news domain; (3) contrastive learning between Neutral and polar pairs to sharpen the decision boundary.

Lexicon methods. VADER (38.6%) and TextBlob (32.3%) fall below the 52.8% majority-class baseline, confirming that assigning the most frequent class outperforms both lexicon systems. The failure stems from the mismatch between

news headline register and social-media-origin lexicons. These methods must not be used for production news sentiment pipelines.

Label construction validity. Our category-based label mapping assumes a monotonic relationship between editorial category and headline sentiment. This is a principled but imperfect proxy: POLITICS headlines can be positive (policy wins) and WELLNESS headlines can be negative (disease outbreak). Future work should validate via human re-annotation of a random stratified sample.

VII. Conclusion

We presented a systematic 5-model benchmark for 3-class news headline sentiment on 204,651 HuffPost headlines. DistilBERT fine-tuned achieves the best practical balance: 80.4% accuracy, macro-F1 0.753, ROC-AUC 0.919, and 3.54 ms/headline GPU inference. BERT-base achieves 81.1% accuracy (F1=0.768) at 5$\times$ the inference cost. Key findings: (1) Lexicon methods collapse on news headlines (VADER 38.6%, TextBlob 32.3%), both below the 52.8% majority-class baseline; (2) the Neutral class remains the hardest across all neural models (best F1=0.63); (3) category difficulty correlates with the reporting/opining spectrum; (4) DistilBERT offers the best accuracy-per-parameter ratio among transformer models; (5) full encoder fine-tuning outperforms frozen training by 6.0 pp; (6) 3.1 pp degradation on live RSS data reveals temporal distribution shift as the dominant deployment challenge. Future work will explore category-conditioned sentiment modeling and contrastive pre-training on headline polarity pairs.

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